

Notes: Gentzkow and Shapiro (JPE, 2006)
Media Bias and Reputation

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1 Big picture

- **Question.** Why do media outlets slant news, even when consumers want accurate information and firms want profits rather than ideology? The paper asks whether reputation concerns alone can generate media bias.
- **Core mechanism.** A Bayesian consumer who is uncertain about a source's quality infers that a source is higher quality when its reports conform to the consumer's prior beliefs. This gives a news firm an incentive to shade reports toward audience priors in order to appear accurate.
- **Bias without taste for bias.** Consumers do not need to enjoy confirmatory information, reporters do not need ideological preferences, and politicians do not need to capture the press. Bias can arise from a rational reputational incentive.
- **Main theoretical result 1: prior-congruent slant.** A normal-quality outlet sometimes suppresses or distorts signals that contradict consumers' priors. Stronger priors imply stronger equilibrium bias.
- **Main theoretical result 2: feedback reduces bias.** If the true state will later be observed, misleading reports risk damaging the outlet's reputation. Bias is therefore lower in high-feedback settings such as weather, sports, or easily verifiable events.
- **Main theoretical result 3: independent competition reduces bias.** A competing outlet can expose a distorted report. More independently owned outlets increase the chance that false slant is revealed and therefore discipline the firm with the scoop.
- **Joint ownership qualification.** If outlets are jointly owned, their incentive is to protect the common reputation. Additional outlets then may merely confirm the original report, so the disciplining effect of competition can disappear.
- **Empirical support.** The paper provides survey, newspaper endorsement, sports forecasting, and local TV evidence consistent with the model: priors shape perceived source credibility, media slant aligns with audience priors, feedback limits bias, and competition is associated with less bias.
- **Takeaway.** Media bias can be an endogenous market outcome of reputation building. The key policy variables are not only ownership ideology but also verifiability and independent competition.

2 Incremental contribution of the paper

- **A reputational theory of media bias.** Earlier theories often attribute bias to consumer tastes, reporter ideology, or government pressure. This paper shows that slant can emerge even when everyone values truth.
- **Bayesian foundation for “telling audiences what they expect.”** The paper formalizes why a rational consumer may infer low quality from a surprising report: inaccurate sources are more likely to report things that contradict a strong prior.
- **Clear comparative statics.** The model predicts where bias is likely to be severe: topics with strong audience priors, weak ex post feedback, and limited independent competition.

- **Competition mechanism distinct from preference-based models.** In some preference or confirmatory-demand models, competition can intensify slant by matching outlets to audience tastes. Here, independent competition can reduce slant by making false reports easier to expose.
- **Ownership implication.** The model links media consolidation to bias through a concrete channel: independently owned outlets check one another, but jointly owned outlets may have incentives to coordinate or confirm.
- **Theoretical and empirical bridge.** The paper combines a formal sender-receiver/reputation model with several empirical exercises, making the mechanism empirically disciplined even though the paper is mainly theoretical.
- **Policy relevance.** The results speak directly to FCC ownership debates and public diplomacy: independent voices and alternative information sources can discipline existing outlets.

3 Data, evidence, and measurement

3.1 Examples of media slant

- The paper opens with three accounts of the same battle in Samarra, Iraq: Fox News, the *New York Times*, and AlJazeera.net. The same underlying event is framed differently through word choice, selective emphasis, and differential credibility assigned to sources.
- This motivates the paper’s definition of media bias: not necessarily fabricating facts, but choosing how to emphasize, omit, and interpret them.
- The model is designed to cover both literal lying and subtler forms of spin. A report can be thought of as choosing whether to emphasize one interpretation or another.

Interpretation: The empirical target is slant in the presentation of information, not merely factual mistakes.

3.2 Evidence that reputation matters

- Media firms incur large costs to gather and verify information, suggesting that credibility is economically valuable.
- The paper discusses major reputational shocks such as CBS’s 2004 National Guard documents story and the *New York Times* Jayson Blair scandal.
- The response to such scandals - apologies, investigations, and resignations - supports the assumption that media firms care about being perceived as accurate.

Interpretation: Reputation is not a decorative feature of the model. It is the payoff-relevant state variable that creates the incentive to slant reports toward priors.

3.3 Evidence that priors affect quality assessments

- Psychological evidence shows that people rate evidence as more convincing when it supports their prior beliefs.

- In the Pew news believability data, conservatives are more likely than liberals to say that they believe Fox News; liberals are more likely than conservatives to say that they believe NPR.
- The paper interprets this pattern as consistent with Bayesian updating about source quality: reports that match priors are more likely to be viewed as coming from a high-quality source.

Interpretation: The paper does not require consumers to irrationally love confirmatory information. A Bayesian quality-inference channel is enough.

3.4 Evidence that reports align with audience priors

- The paper discusses anecdotal evidence from southern newspaper coverage of lynchings, Arab versus American coverage of the Iraq War, and CNN domestic versus CNN International coverage.
- It also presents evidence from 2000 newspaper endorsements: endorsements for George W. Bush are more common in lower-income states, paralleling voting patterns.
- These patterns suggest that media outlets often tailor slant to the beliefs of their audiences.

Interpretation: The audience-prior channel is visible both across countries and within the United States.

3.5 Empirical checks on feedback and competition

- To examine feedback, the paper uses sports prediction data from *New York Times* editors, where outcomes are quickly observed. The model predicts limited bias in this environment.
- To examine competition, the paper uses 2000 Local News Archive data on local TV coverage of Bush and Gore.
- The bias measure is based on the imbalance in seconds of speaking time given to Bush and Gore by each station.
- The key explanatory variable is the number of local news broadcasts in a media market.

4 Models and theoretical specifications

4.1 Consumer decision problem

There is a binary state of the world,

$$S \in \{L, R\}, \tag{1}$$

and consumers choose a binary action,

$$A \in \{L, R\}. \tag{2}$$

Consumers receive payoff 1 if $A = S$ and 0 otherwise. Consumers have prior probability v that the state is R .

Interpretation: News is valuable because it helps consumers choose the correct action. The model abstracts from direct entertainment value or ideological consumption utility.

4.2 Firm types and signals

A media firm can be one of two quality types:

- with probability λ , it is high quality and observes the true state perfectly;
- with probability $1 - \lambda$, it is normal quality and observes an imperfect signal $s \in \{l, r\}$.

For a normal firm,

$$\Pr(l \mid L) = \Pr(r \mid R) = p, \quad p \in (1/2, 1). \quad (3)$$

High-quality firms report honestly. Normal firms can report either $\hat{l}$ or $\hat{r}$ regardless of their signal.

Interpretation: The normal firm has discretion over spin. The high-quality type anchors the reputational incentive: imitating a high-quality source is valuable.

4.3 Reputation payoff

After the report, and possibly after feedback about the true state, consumers update their belief that the firm is high quality. Let

$$\lambda(\hat{s}, X) \quad (4)$$

be the posterior probability that the firm is high quality after report $\hat{s}$ and feedback $X \in \{L, R, 0\}$, where $X = 0$ denotes no feedback.

The firm receives continuation payoff

$$\sum_i f(\lambda_i), \quad (5)$$

where f is strictly increasing.

Interpretation: The media firm cares about future demand for its reports. The current report is chosen partly to preserve or improve perceived quality.

4.4 No-feedback intuition

When there is no feedback, a report that matches consumers' prior is interpreted as stronger evidence of quality. If consumers strongly expect R , then a report $\hat{r}$ makes the firm look more accurate than a report $\hat{l}$.

In equilibrium, the normal firm must sometimes distort signals away from truth and toward consumer priors. With $v > 1/2$, the normal firm sometimes reports $\hat{r}$ after observing l .

Interpretation: Bias is not driven by the firm's belief that R is true. It is driven by the consumer's inference that an R -consistent report is more likely to come from a high-quality source.

4.5 Feedback and truthful reporting

Suppose consumers learn the true state with probability μ . If the firm slants a report and the truth is revealed, the mismatch damages reputation. The expected gain from slanting falls as μ rises.

Proposition 1 states that the unique equilibrium strategy for a normal firm is:

- report r signals truthfully;

- misrepresent l signals as $\hat{r}$ with probability $j_l^*(\hat{r}) \in [0, 1]$.

The distortion probability is weakly increasing in the consumer prior v , and, when positive, strictly decreasing in feedback probability μ .

Interpretation: Bias increases with audience certainty and decreases with verifiability.

4.6 Welfare

Because bias garbles information, eliminating bias can make consumers weakly better off and can increase current firm profits when the prior probability of high quality is small. This result highlights that equilibrium bias can be privately rational but socially inefficient.

Interpretation: Bias is not a Coasian matching of consumers with desired news. It can be a distortion that reduces information quality for all market participants.

4.7 Independent competition

The paper extends the model to J outlets. One firm has the scoop, and the other firms can provide feedback. If outlets are independently owned, feedback firms report the truth to protect their own reputations. More outlets increase the probability that consumers see corrective feedback.

Proposition 3 states that equilibrium bias is weakly decreasing in the number of firms J , and strictly decreasing whenever bias is positive. For sufficiently large J , bias falls to zero.

Interpretation: Competition works because independent outlets can expose each other's inaccurate reports.

4.8 Joint ownership

If all outlets are jointly owned and share a common reputation, feedback outlets have an incentive to confirm the scoop firm's report rather than expose it. Proposition 4 shows that under joint ownership, equilibrium bias is independent of the number of outlets.

Interpretation: Ownership independence matters. The same number of media outlets can have very different effects on bias depending on whether they are separately controlled.

4.9 Heterogeneous priors and market segmentation

The model also allows consumers with different priors to coexist in one market. Suppose one consumer group has a prior favoring R and another has a prior favoring L . Then there can be a segmented equilibrium:

- one firm serves the R -prior group and biases toward $\hat{r}$;
- another firm serves the L -prior group and biases toward $\hat{l}$.

There are also equilibria with honest reporting when each outlet serves a balanced audience.

Interpretation: Segmentation by priors can sustain persistent ideological differentiation across outlets.

5 Identification and empirical design

5.1 What the theory identifies

- The theoretical model identifies a mechanism: reputation plus Bayesian inference can produce slant toward audience priors.
- The mechanism does not rely on consumer demand for enjoyable bias, journalist ideology, or direct political pressure.
- The model generates comparative statics rather than a single estimated structural parameter: stronger priors increase bias, feedback reduces bias, and independent competition reduces bias.

Interpretation: The central identification is conceptual. The paper isolates a mechanism that can rationalize media slant without ideological preferences.

5.2 Empirical evidence and limits

- The empirical exercises are supportive rather than definitive causal tests.
- The Pew evidence shows that perceived credibility varies with political priors, but it does not prove that quality inference is the only channel.
- The endorsement evidence shows that newspaper slant aligns with local political tendencies, but local politics may also correlate with owner ideology, advertiser demand, or candidate strategy.
- The sports evidence is a useful high-feedback comparison, but it does not directly test all dimensions of political reporting.
- The local TV evidence links competition to less Bush-Gore coverage imbalance, but the number of local broadcasts may still correlate with unobserved market characteristics.

Interpretation: The evidence is best read as a collection of patterns consistent with the model's comparative statics.

5.3 Alternative explanations

- **Consumer taste for confirmation.** Consumers may prefer news that flatters their beliefs. The paper argues that its Bayesian reputation channel can produce similar facts without assuming such tastes.
- **Reporter ideology.** Journalists may have ideological preferences. The paper does not deny this, but shows that bias can arise even without it.
- **Political pressure.** Governments and parties can influence media. The model instead emphasizes market-based incentives, though political pressure can interact with the competition mechanism.
- **Audience sorting.** Different audiences may sort into outlets that match their views. The heterogeneous-priors section explains how such segmentation can be sustained endogenously.

6 Tables and figures

6.1 Figures 1 and 2: Priors, source credibility, and audience-congruent slant

Read:

- Figure 1 uses Pew survey data. Conservatives are more likely to say that they believe Fox News, while liberals are more likely to say that they believe NPR.
- The pattern is monotone across political views: as respondents become more liberal, belief in NPR rises and belief in Fox News falls.
- Figure 2 compares Bush's vote share across state income quartiles with newspaper endorsements for Bush.
- Newspapers endorse Bush more often in states where Bush receives more votes, suggesting that editorial positions align with local audience beliefs.

Economic message: The first figure supports the model's assumption that priors shape perceived source quality. The second figure supports the idea that media reports and editorial positions tend to move toward audience priors. Together, they motivate the reputational slant mechanism.

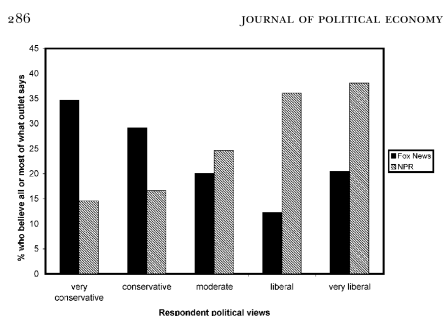


FIG. 1.—Political views and assessments of news media believability. Data come from the Pew Research Center for the People and the Press's 2002 News Media Believability Survey. Exact wording for a survey question on respondent political views: "In general, would you

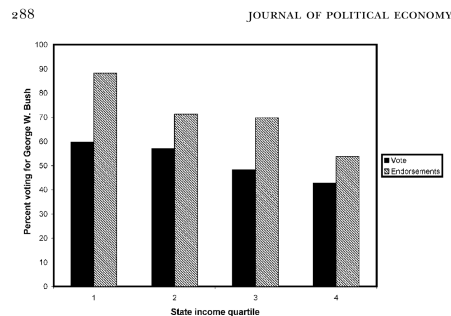


FIG. 2.—Newspaper endorsements and ideology across U.S. states in the 2000 election. Data on voting behavior are taken from the Federal Election Commission (<http://www.fec.gov/elections.html>). Percent for Bush reflects percentage of the two-party vote. Data

(a) Figure 1: political views and media believability

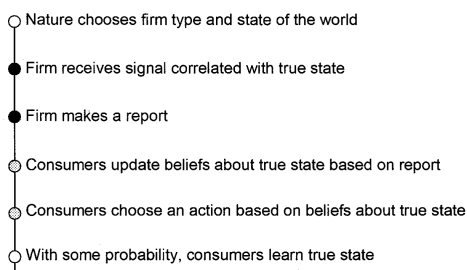
(b) Figure 2: endorsements and state ideology

6.2 Figures 3 and 4: Timing of the model and comparative statics

Read:

- Figure 3 lays out the monopoly game: nature chooses type and state, the firm observes a signal, the firm reports, consumers act, feedback may arrive, consumers update source-quality beliefs, and the firm receives a continuation payoff.
- The timing makes clear why reports affect future payoffs: even after consumers choose an action, they may learn whether the report matched the state.
- Figure 4 plots the equilibrium degree of bias against the consumer's prior belief.
- Bias rises as priors become stronger. The dashed no-feedback curve is above the feedback curve, showing that feedback disciplines distortion.

Economic message: The model's timing turns reputation into an incentive constraint. The numerical example visualizes the paper's core comparative statics: bias is high when audiences have strong priors and low when reports are likely to be checked.



(a) Figure 3: timing of the monopoly game

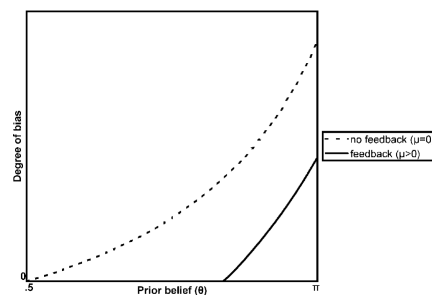


FIG. 4.—Numerical example. Vertical axis shows equilibrium probability of reporting $\hat{\tau}$ given that the newspaper with the scoop receives a signal of $\bar{L}$. Horizontal axis shows prior probability θ . Dashed line shows equilibria for the limit case of no feedback ($\mu = 0$).

(b) Figure 4: bias, priors, and feedback

6.3 Figure 5 and Table 1: Feedback and competition in empirical checks

Read:

- Figure 5 studies *New York Times* sports picks. Since game outcomes are quickly observed, the model predicts limited slant toward local teams.
- The figure does not show New York teams as the uniquely favored teams, consistent with feedback reducing incentives to pander.
- Table 1 studies local TV news coverage of the 2000 election.
- The dependent variable measures imbalance in speaking time between Bush and Gore. A larger number of local news broadcasts is associated with lower bias.
- The negative coefficient on the number of local news broadcasts remains similar after controls for region, population, and income.

Economic message: The sports evidence is consistent with the idea that fast verification limits bias. The TV evidence is consistent with the competition prediction: more independent news sources reduce the profitability of one outlet's slant because competitors can reveal contrary information.

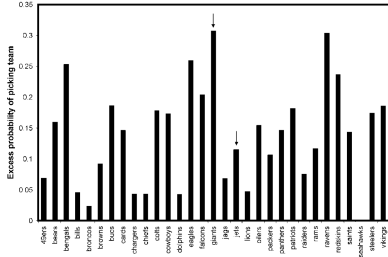


FIG. 5.—Sports picking by *New York Times* sports editor, 1994–2000. Data come from Boulter and Sieckler (2003). The data set contains information on the picks of the *New*

(a) Figure 5: sports picking and high-feedback reporting

TABLE 1
COMPETITION AND BIAS IN LOCAL NEWS COVERAGE OF THE 2000 ELECTION

	(1)	(2)	(3)	(4)
Number of local news broadcasts, 2002	-.0057 (.0025)	-.0064 (.0025)	-.0062 (.0030)	-.0062 (.0031)
Census region controls?	no	yes	yes	yes
Log(population), 2000			-.0006 (.0043)	-.0004 (.0053)
Log(income per capita), 1999				-.0013 (.0188)
Observations	58	58	58	58
R ²	.0834	.1747	.1751	.1752

NOTE.—Bias is measured as the average across all stations in a market of

$$\text{bias}_i = \left(\frac{\text{bush}_i}{\text{bush}_i + \text{gore}_i} - \frac{1}{2} \right)^2$$

(b) Table 1: competition and local-news bias

7 Short summary

- The paper develops a reputational model of media bias.
- Consumers are Bayesian and value accuracy, but they use reports to infer source quality.
- Reports that match consumers' priors make an outlet look more reliable.
- A normal-quality outlet therefore sometimes slants information toward the audience's prior beliefs.
- Bias is stronger when consumers have stronger priors.
- Bias is weaker when the true state is likely to be revealed later.
- Independently owned competitors reduce bias by increasing the chance that distorted reports are exposed.
- Jointly owned outlets may not discipline each other, so consolidation can weaken the competition channel.
- Heterogeneous priors can generate market segmentation, with different outlets slanting in different directions.
- The empirical evidence is broadly consistent with the model: priors shape believability, endorsements align with audience views, sports forecasts show limited local pandering, and local TV bias is lower in more competitive markets.

8 Conclusion

8.1 Core conclusion

Media bias can arise from reputation-building incentives alone. A firm that wants to look accurate may avoid reports that contradict its audience's prior beliefs, because consumers rationally interpret surprising reports as possible evidence of poor source quality.

8.2 Economic intuition

The model turns a simple Bayesian logic into a market distortion. If consumers are confident that a state is likely, then an outlet reporting the opposite state looks less competent. A profit-maximizing outlet internalizes this inference and tilts reports toward the prior. The result is a loss of information even though consumers value truth.

8.3 Incremental contribution revisited

The paper’s main contribution is to separate media bias from ideology or taste-based demand for slant. It shows that even a truth-oriented consumer market can produce biased equilibrium reporting when source quality is uncertain and reputation matters.

8.4 Implications for media markets and policy

- Media ownership rules matter because independent outlets can check one another’s reports.
- More outlets do not automatically imply less bias if ownership is consolidated.
- Public diplomacy may be more effective when it supports alternative independent information sources rather than simply pressuring existing outlets.
- Topics with delayed or ambiguous feedback - wars, taxes, climate change, foreign policy - are likely to be especially vulnerable to reputational slant.

8.5 Limitations

- The empirical evidence is suggestive, not a fully identified causal estimate of the model’s parameters.
- The model abstracts from many real-world features: advertising markets, platform algorithms, reporter ideology, ownership preferences, and direct political pressure.
- High-quality firms are assumed to report truthfully, and normal firms’ reporting discretion is simplified.
- The model focuses on source reputation, not on persuasion, entertainment, or identity utility from news consumption.

8.6 Takeaway

The central lesson is that bias can be a reputational equilibrium. When audiences judge quality partly by whether reports sound plausible relative to what they already believe, profit-maximizing media firms have incentives to shade information toward those beliefs. Feedback and independent competition are the main disciplining forces.